APMC Mandi Market Intelligence System

District-Level Agricultural Market Condition Classification — Project Overview

1. Project Direction

The project will focus on agricultural market intelligence at the district level. The current dataset is coherent with the overall idea because it contains market, commodity, variety, date, and minimum/maximum/modal price information. The project should be narrowed to predicting future market conditions rather than trying to predict an exact price.

2. Dataset Fit

- **Available:** State, District, Market, Commodity, Variety, Arrival Date, Minimum Price, Maximum Price, Modal Price.
- **Dataset size:** approximately 9,558 records and 11 columns.
- **Important limitation:** the current dataset does not contain reliable arrival-quantity information.
- **Therefore:** arrival-quantity features should be removed unless a separate reliable source is integrated.
- **Data limitation:** approximately one year of data may not capture long-term seasonal patterns.

3. District-Level Analysis

The project should be described as district-level market intelligence, while retaining mandi-level information as an input. A suitable hierarchy is:

District → Commodity → Variety → Market/Mandi → Historical Price Behaviour

Instead of simply combining every record by district, the system can use individual mandi information and create district-level aggregated features. This preserves useful market information while keeping the final scope at district level.

4. Recommended Feature Engineering

- Historical modal price / previous price.
- Recent price averages such as 3-day and 7-day averages, where observations permit.
- Recent price change and price trend.
- Price volatility.
- Month, week, season, and other date-based features where appropriate.
- District-level average/summary price across relevant mandis.
- Neighbouring-mandi price information, such as average nearby price and target-vs-neighbour price difference.
- Weather features only if reliable historical weather data can be matched without leakage.

5. Remove Regression?

Yes. Regression can be removed and the project can become a classification-only ML project. This makes the objective more focused: instead of predicting an exact future modal price, the model predicts the expected future market condition.

6. Classification Target

The proposed classes are **Favorable, Normal, and Unfavorable**. The classes must be created using a clearly documented, data-based definition based on future price behaviour or an appropriate benchmark. The thresholds should be checked against the actual dataset so that the classes are sufficiently represented.

7. Classification Models

- Logistic Regression
- Decision Tree Classifier

- Random Forest Classifier
- Gradient Boosting Classifier

8. Evaluation

Since regression is removed, MAE, RMSE, and R^2 are no longer required. The classification system should be evaluated using Accuracy, Precision, Recall, F1-score, and a Confusion Matrix. Class imbalance should also be checked before relying on accuracy alone.

9. Neighbouring Mandi Intelligence

This can remain as a distinctive part of the project. The purpose is to test whether nearby market behaviour provides useful information for classifying the target market/district condition. Possible features include nearby average price, price difference, price trend, and the proportion of nearby markets showing increasing or decreasing prices. Only information available at prediction time should be used.

10. Data Leakage Prevention

- For a prediction made at time T, inputs must contain only information available by T.
- Do not use future modal prices as input features.
- Do not use future weather or future market information.
- Construct the classification target from future behaviour, but keep that future information completely separate from model inputs.
- Use chronological train/validation/test splits rather than randomly shuffling the time series.

11. Revised Project Workflow

Government mandi data → Data verification → Cleaning → EDA → Feature engineering → District/mandi aggregation → Classification target creation → Chronological train/validation/test split → Classification models → Evaluation → Feature contribution analysis → Final market intelligence output

12. Final System Output

- Selected district.
- Commodity and variety.
- Relevant mandi/market information.
- Expected market condition: Favorable / Normal / Unfavorable.
- Supporting recent price trend information.
- Neighbouring-mandi summary where available.

13. Key Changes from the Current Proposal

Area	Current proposal	Recommended version
Main ML tasks	Regression + Classification	Classification only
Target	Future modal price + condition	Future market condition
Regression metrics	MAE, RMSE, R^2	Remove
Classification metrics	Accuracy, Precision, Recall, F1	Keep + Confusion Matrix
Arrival quantity	Planned if available	Remove from current dataset unless added reliably
Scope	Lucknow district	District-level market intelligence
Mandi data	Market/Commodity/Variety series	Use mandi information as inputs, produce district-level analysis

14. Final Recommended Project Statement

We are developing a district-level APMC Mandi Market Intelligence System that uses historical agricultural market data to classify the expected future market condition of selected commodities. The system will use historical modal prices, seasonal information, mandi-level information, and neighbouring-market intelligence as predictive features. Classification models will categorize future conditions as Favorable, Normal, or Unfavorable. The models will be evaluated using chronological validation and classification metrics, while data leakage and class imbalance will be carefully addressed.